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Heatwave-related mortality among older adults in Sweden: A nationwide case-crossover study on the roles of sex and living environments

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Abstract

This is a placeholder abstract of the thesis. It summarises the study, the methods used, and the main results in a concise form. Maximum 300 words.

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A. Popular Science Summary **24**

1 Introduction

1.1 Background

Heatwaves have increased, and are projected to increase, in frequency, magnitude, duration, and spatial extent (1). Heat stress is the leading cause of weather-related deaths (2), and consequently, heatwaves are considered to be one of the most dangerous weather hazards (3). In the Nordic countries, heatwaves are expected to become five times more frequent by the end of the century under current climate trajectories (4), making the health consequences of prolonged periods of heat an urgent concern for susceptible groups living in Sweden.

When exposed to abnormal heat, the human body attempts to cool itself in order to maintain normal physiological functioning. This process, known as thermoregulation, allows the body to control its internal temperature despite varying environmental conditions (5). It increases its heart rate to raise cardiac output, the total volume of blood pumped by the heart's ventricles. This redistributes blood to the skin, where heat is lost (1,5). Simultaneously, it increases sweating (5). Thermoregulation puts most stress on organs of the circulatory and respiratory system, namely the heart, lungs, and kidneys (2). If uncapable of matching heat absorption with heat loss, the body experiences overheating, or hyperthermia. Its consequences range from mild, such as dizziness, flushed skin, or fainting, to more severe, such as heat exhaustion and heat stroke, which can be fatal. Exposure to extreme heat is especially dangerous to those living with chronic conditions affecting the circulatory and respiratory systems, as their bodies have to withstand additional physiological stress (2).

Adults aged 65 years and above represent the fastest growing demographic group in Sweden (6), and are especially vulnerable to heatwaves due to a higher prevalence of chronic conditions, limited capacity for thermoregulation, as well as age-related bodily decline (7). Physiological scholarship has shown that older adults are less tolerant to heat than younger adults as they tend to experience greater heat storage, faster increase in core temperature, and are at higher risk of dehydration (7). This susceptibility is also reflected in epidemiological research, that consistently finds excess and heat-attributable mortality among older adults during heatwaves (8–12). These estimates have risen substantially through recent decades. A 2023 Lancet report estimated that the increase in heatwaves has caused a 85% rise in heat-related mortality for people over 65 years of age between 2000–2004 and 2017–2021 worldwide (13).

Nevertheless, older adults are a highly heterogenous group, also in regard to heat suscept-

ibility. With advancing age, vulnerability increases (6,14,15), with, in some contexts, adults being 85 years and above to have been found to be the most at-risk age group (16,17). Because individuals acclimatise to their usual climate and are susceptible to local extreme temperatures, older adults in Sweden, despite living in a cold climate, are not safe from heatwave-related risks (ref.).

According to the World Health Organisation (WHO), the European continent is the fastest-warming region with temperatures rising at approximately twice the global rate (18). On top of that, in the European context, the Nordic countries are among the most affected areas due to the phenomenon known as Arctic amplification, whereby the Arctic regions have warmed nearly four times faster than the global average since 1979 (19). This accelerated warming is primarily driven by the diminishment of snow and sea ice, and changes in ocean and air circulation (19). The warming of the Arctic contributes to longer summer weather persistence, including prolonged periods of extreme heat in Sweden (20). Hence, temperatures in Sweden are expected to rise even more sharply than in the rest of Europe, and, if adaptive climate strategies are not implemented, this is likely to continue to result in higher mortality rates during heatwaves. A 2020 study, examining Swedish weather data of warm summer months between 1990 to 2014, found that heatwaves are associated with a 10% increase in all-cause mortality (21). Similar increases have been observed in other Nordic countries (22,23).

That said, although an extensive body of literature exists regarding temperature-related mortality, there is a notable scarcity of studies on the topic carried out in Scandinavian settings (8). Swedish houses are generally unsuited to withstand hot weather, as they are primarily tailored to winter climates. They are commonly well insulated, to save heating and prioritise thermal comfort. However, they are rarely designed to manage overheating, which can lead to unbearably high indoor temperatures during heat events (24). Understanding the role of living environments during heatwaves therefore becomes of pivotal importance, as they can mitigate or exacerbate vulnerability to heat-related health risks.

Living environments are a well-established, yet often overlooked (25,26), determinants of health for older adults (27,28). They encompass both housing characteristics, such as housing conditions, quality, and accessibility, as well as residential aspects, such as neighbourhood deprivation and urbanicity status (29). As people grow older, enter retirement, or experience physical limitations, they tend to spend more time at home and within their neighbourhoods. Therefore, the features of our living environments become increasingly important for health as

we age (26). In the context of heatwaves, living environment characteristics, such as insulation, access to cooling, urbanicity status, as well as proximity to green spaces, can mediate the degree of heat exposure and therefore shape health outcomes (30). Despite this, neither in Sweden, nor in any other Nordic country, has there been any research conducted to investigate how housing and residential characteristics influence heatwave-related health outcomes among older adults. A systematic review published in 2025 found that the majority of studies on the topic have been carried out in Northern American settings, whilst few were set in European countries, and none in the Nordics (31).

Similarly, the role that pre-existing health conditions and sex play in heat-related health outcomes is not yet fully understood. While the association between heat and all-cause mortality internationally well-studied, limited studies assess its associations to cause-specific mortality (32) and findings are often less consistent (32,33). Further, when cause-specific mortality is taken into consideration, most epidemiological studies tend to focus on broad outcome categories, namely respiratory and cardiovascular outcomes, often classified according to the chapters subdivision of the International Classification of Disease (ICD) (34). Using these broad health outcomes categories, although statistically convenient, results in two constraints. Firstly, a broad classification of outcomes limits our understanding of which individuals are most at risk, and secondly, it does not allow to identify whether specific diagnoses are driving observed associations. Thus, in many cases, how specific diagnoses impact heat-related mortality remains unexplored (33,35).

With regard to sex differences in heat-related health outcomes, studies carried out in Sweden do not align with international findings. On the international level, three recent systematic reviews have observed that the majority of studies find that women are more susceptible to heatwave- and heat-related mortality. A 2025 study set in urban Norway found similar associations, with older women being more at risk during heatwaves (23). Nevertheless, evidence remains heterogenous, and sometimes inconsistent, as findings differ based on the definition of exposure, population, and setting (36). Literature from Sweden, which has predominantly been carried out only in Stockholm and Stockholm county (21), has, in a few instances, found men above the age of 65 to be at higher vulnerability than women (37,38). That said, generally in Swedish studies no significant difference is observed between the two sexes. This is also the case for older adults; although men were found to have a slightly higher mortality risk than women, this difference was not statistically significant (37).

In summary, it is evident there is still much to be learned about which factors shape vulnerability to heatwaves among older adults living in Sweden. Considering that housing policies, building stock, and population vulnerability to heatwaves are inherently context-specific, the absence of evidence from the local and national Swedish levels represents a critical gap, one that limits the capacity of policymakers to develop targeted mitigation strategies to promote the creation of age-friendly living environments in the context of heatwaves.

1.2 Aim

The present study seeks to contribute to this body of literature by linking nationwide individual-level data on sociodemographic factors, living environments, mortality, and heatwave exposure of older adults, aged 70 years and above, living in Sweden spanning 2014 to 2018. Its primary aim is to determine whether sex, age group (70–84 and 85 years and above), and living environment characteristics act as effect modifiers in the association between heatwaves and mortality. Specifically, the study investigates how year of construction, housing type, and urbanicity influence heatwave-related all-cause mortality across age groups, and whether these associations differ between men and women. Additionally, this study further stratifies results by underlying cause of death to explore potential differences in susceptibility among older adults.

1.3 Research Questions

1. Do age group (70–84 and 85+), sex, and specific living environment characteristics, namely year of construction, housing type, and urbanicity, act as effect modifiers on the association between heatwaves and mortality among older adults in Sweden?

2. How does heatwave-related mortality differ across underlying causes of death, and do these patterns vary between age group?

2 Methods

2.1 Study design

The study employs a case-crossover design, where participants serve as both cases and controls. Exposure windows of the day the event takes place and at other control times are compared. In this study, whether a heatwave occurred on the day of death (case) is compared with non-event days (controls). This eliminates time-invariant confounding, as participants are compared to

themselves (39). This design is apt for studying the association between sudden outcomes, such as heat-related mortality, and short-term or transient exposures, such as heatwaves, and is readily used to measure the association between heat exposure and health outcomes (40). Furthermore, this approach is particularly powerful in case of weather-related exposures because it allows to analyse large numbers of cases and use routine weather data to assess exposure (39).

2.2 Study setting, period, and population

This study is set in Sweden and analyses data from April to September over the years 2014 to 2018, consistent with Swedish meteorological definitions and prior seasonal epidemiological research. According to the Swedish Meteorological and Hydrological Institute (SMHI), spring usually arrives in most of Sweden by April, while summer temperatures usually linger well into September (41). Epidemiological studies in Sweden and abroad frequently restrict analyses to warm months when investigating heatwave-related outcomes, since heatwaves are inherently seasonal exposures (21,42). Therefore, to capture the full extent of heatwave exposure, the study period was defined as April to September over each year.

The study population was limited to individuals aged 70 years and older who died during the study period. This age threshold was established for two reasons. First, heatwave exposure was assigned to individuals by their home residence. Considering retirement age in Sweden typically lingers between 65 to 69, it is reasonable to assume that individuals aged 70 and above are retired and therefore present at their residence address throughout the day, minimising exposure misclassification. Secondly, this group has consistently been found to be particularly vulnerable to heat-related mortality (43), and to cardiovascular and circulatory diseases (44) than younger population cohorts.

2.3 Data Sources

All individual level data was obtained from Register RELOC-AGE. Register RELOC-AGE is a general population register-based cohort in Sweden with yearly data available between 1987 and 2021. Through the linkage of multiple population-based registers, it offers high quality data on health outcomes, including mortality data, sociodemographic and socioeconomic characteristics, housing, and neighbourhood data for individuals aged $\geq$ 55 years (45).

2.3.1 Mortality data

The Cause of Death Register provided by the National Board of Health and Welfare was used to identify the participants of the study. Daily counts of mortality due to all causes, as well as due Cardiovascular Disease (CVD) (ICD-9: 390–459; ICD-10: I00–I99), Coronary Heart Disease (CHD) (ICD-9: 410–414; ICD-10: I20–I25), and Myocardial Infraction (MI) (ICD-9: 410; ICD-10: I21) were analysed.

2.3.2 Residential and housing data

Residential coordinates and housing characteristics were obtained from the Swedish Real Estate Property Register (REPR) and the Apartment Register (AR), both maintained by Statistics Sweden. As moves to residential care facilities were not consistently captured in the REPR, supplementary information on residential care facility residence was obtained from the National Register of Care and Social Services for the Elderly and Persons with Impairments (NRCSS), maintained by the National Board of Health and Welfare. Only individuals with residential coordinates recorded in 2017 or 2018 were eligible for inclusion.

2.3.3 Demographic and socioeconomic data

Demographic and socioeconomic characteristics were retrieved from the Longitudinal Integrated Database for Health Insurance and Labour Market Studies (LISA), maintained by Statistics Sweden. Disposable income was divided into quintiles using annual equivalised household income.

2.3.4 Air temperature data

Data on daily maximum air temperatures (TX) was collected from the openly accessible SMHI's Gridded Climatology (46). The raster dataset covers the territory covering the Nordic countries on a grid with 2.5 km^2 horizontal resolution using Lambert Conformal Conic projection. A daily measure of TX is attributed to each grid cell from January 1st, 1961, up to and including December 31st, 2018 (46). The data was cropped to cover Swedish territory only, which consists of 74,862 cells, each covering 2.5 km^2. The area covered is therefore of 467,887.5 km^2. Swedish territory, according to Nordic Statistics accounted for 447,428 km^2 in 2024 (47). The area covered by the cropped gridded dataset is slightly larger as the grid cells that follow the coastline

also include the airspace beyond the border. Although minimum daily temperature (TN) is also readily used as metric to calculate heatwave thresholds (48), I preferred using TX as previously literature notes that this metric has been better at capturing mortality than TN (49).

2.3.5 Urbanicity data

The Nordic urban-rural typology was used to assess urbanicity. The data covers the Nordic countries and uses a grid-based classification of areas based on population density, proximity measures and land cover parameters to establish the level of urbanisation. The typology uses seven classes that range from inner urban area to sparsely populated rural area (50). These seven classes were collapsed into a single binary variable and were used to assign whether participants lived in a rural or urban area.

2.4 Heatwave definition

A heatwave is a prolonged period of unusually high temperature. There is no single, internationally accepted definition of a heatwave (51). Definitions vary regionally, in both public health and policy literature, in part because a heatwave is measured relative to the usual weather in the area and relative to normal temperatures for the season, and in part because there is no single best indicator from a public health perspective as different heatwave thresholds will capture different outcomes risks (48,52). In Sweden, according to the SMHI's handbook Väderspråket, a heatwave is defined as a period of at least five consecutive days with a TX of at least 25°C (53). This is a widely adopted meteorological definition in Sweden (51,54) and has been used to issue weather warnings (21). It applies uniformly across Sweden, without accounting for regional variations (21).

It is a well-established issue that this meteorological definition ignores two considerations that are relevant to public health research, making it unsuitable to measure heatwave impacts on health outcomes in epidemiological studies. First is geographical weather variability within Sweden. Air temperatures vary by area, and this is an especially important point to consider given that Sweden covers a vast territory. A weather event seen as severe in one place can be seen as typical in another (51,55). Therefore, magnitude and intensity measured in absolute measurements cannot be compared. Secondly, there is evidence that outlines that heat-related risks are greater in colder climates, suggesting that populations adapt to their usual climate and are more vulnerable to local extreme temperatures (21,56,57). These issues can be resolved by

using a percentile-based definition (51).

Percentiles are a statistical way of describing the position of a value within a distribution, and in heatwave research, they can be used to create location-specific, climate-adjusted temperature thresholds (51). A percentile is calculated from the distribution of historical daily temperatures of a specific location, in this case a 2.5 km^2 grid cell, and when set at a specific threshold, such as 90th, 95th, or 99th percentile, it is possible to define hot or extremely hot days in a comparable and consistent manner (48,51). For example, a 90th percentile of daily TX for a given date and location means that 90% of historical temperatures on that date were below this value, and only 10% were above it, therefore exceeding the 90th percentile means it is hotter than usual for that place and day of the year. To establish these thresholds, one can use a reference period. According to the World Meteorological Organisation, the historical reference period to be used to establish whether temperatures fall within typical range or not should be 'the most recent 30-year period terminating in a year ending with a zero" (58,59).

Per grid cell (g), the 95th percentile temperature threshold (P_{95}) was found by calculating the 95th percentile ($Q_{0.95}$) of all April to September daily maximum temperatures (TX) on days (s) within this reference period. The 95th percentile threshold for each grid cell was calculated as:

$$P_{95g} = Q_{0.95}(TX_{g,s}), \quad s \in [Apr, Sep], \quad y \in [1981, 2010]$$

If the temperature on a day during the observation period (t) was higher than the percentile in the same grid cell, it was considered an exceedance. An exceedance was defined as:

$$E_{g,t} = \begin{cases} 1 & \text{if } TX_{g,t} > P_{95}(g) \\ 0 & \text{otherwise} \end{cases}$$

Previous scholarship notes that it is instrumental to integrate duration and intensity to define heatwaves so to capture health outcomes (60). If exceedances appeared for five consecutive days or more, then the period was considered a heatwave (HW).

$$HW_{g,t} = 1 \text{ if } \sum_{\tau=t}^{t+4} E_{g,\tau} = 5$$

Another three definitions of heatwaves were considered for sensitivity analyses, and their results are attached in the Appendix.

2.5 Exposure assessment and statistical analyses

Within the case-crossover design, a time-stratified approach was adopted to estimate the association between heatwave exposure and mortality. Individual heatwave exposure was determined by linking each participants' residential coordinates to the corresponding grid cell of the heatwave percentile data, and a heatwave indicator variable was extracted for both the case and control days. Assessing heatwave presence through a daily binary variable is a common approach (61). This allowed to test whether there was an association present between heatwaves thresholds and odds of mortality using conditional logistic regression models. By conditioning on the individual, all time-invariant personal characteristics, such as sex and race, are implicitly controlled for.

Case days were determined as the date of death. Control dates were selected using a unidirectional referent selection approach, defined as the same weekday as the case day on the four preceding weeks. This resulted in each case having four controls on the same weekdays prior to their passing. For example, if the case occurred on a Sunday, the control days were the four preceding Sundays (Figure 1). This time-stratified reference selection ensures that control days are drawn from the same seasonal and weekly context as the case day, controlling implicitly for long-term time trends and seasonality (62). Also, because case and control days are separated by exact multiples of seven days, serial autocorrelation in both exposure and outcome is substantially reduced, supporting the independence assumption of conditional likelihood (63). Time-stratified control selection is a common and robust methodological approach used within environmental epidemiology (64–66).

All data management and statistical analyses were performed in R (version 4.5.1; R Core Team, 2025). Conditional logistic regression was conducted using the `clogit` function in the `survival` package (version 3.8-3). Handling of air temperature data and spatial linkage of residential coordinates to raster was performed using the `sf` (version 1.0-21) and `terra` (version

1.8-86) packages.

2.6 Ethical considerations

The Swedish Ethical Review Authority approved the Register RELOC-AGE project and inherent studies (dnr. 2020-01369; 2021-01124; 2024-02460-02). Since this was a register-based study using pseudonymized data, informed consent to participate was not required. Instead, participants can exclude themselves from Register RELOC-AGE by an opt-out message that is available through the Lund University Research Portal (67). The data were stored, managed and analysed on a high-security platform for handling sensitive data (LUSEC) at the Faculty of Medicine, Lund University. Regulations in accordance with the General Data Protection Regulation (GDPR) and ethical principles outlined in the Helsinki Declaration for research involving humans were followed.

3 Results

Study population characteristics are presented in Table 1. The distribution of underlying causes of death is shown in Table 2. Temperature statistics across the study period are summarised in Table 3. Results stratified by living environment and effect modification are shown in Table 4, and results stratified by underlying cause of death are presented in Table 5. The case-crossover study design and control selection are illustrated in Figure 1, with heatwave spatial coverage shown in Figure 2.

4 Discussion

5 References

6 Figures

Figure 1: Illustrative example of time-stratified control day selection. Adapted from Tobias et al.*30.

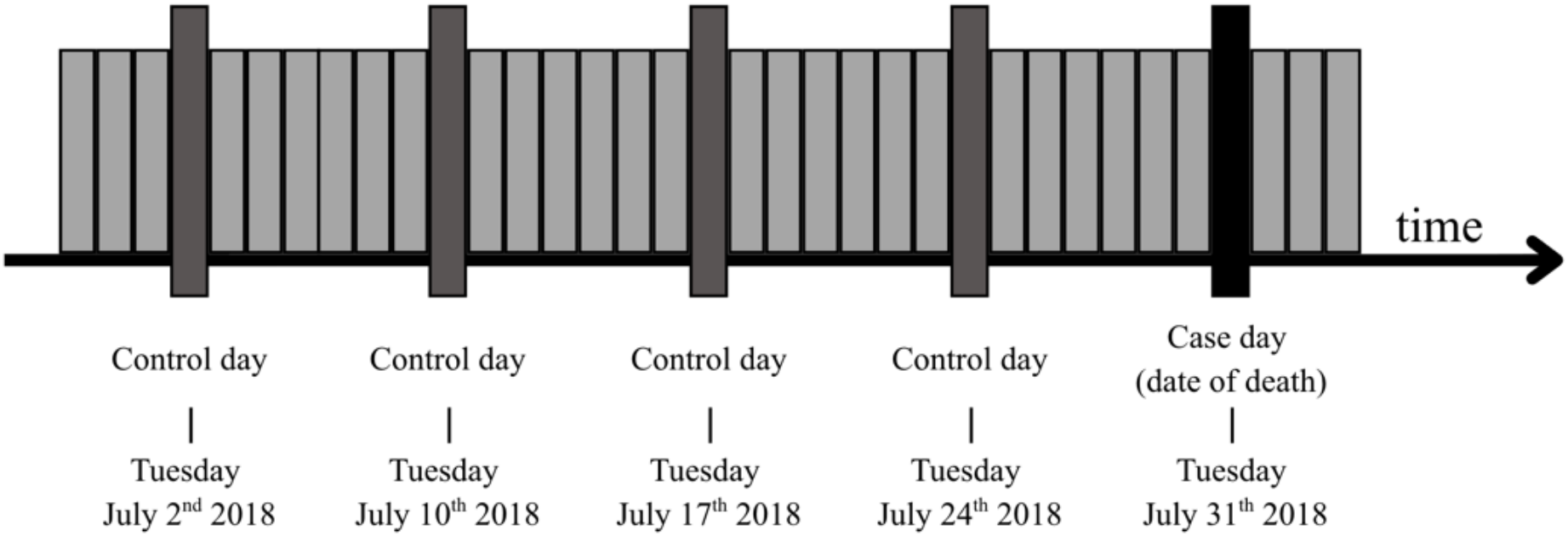


Figure 2: Cumulative heatwaves by municipality (May to September 2014–18)

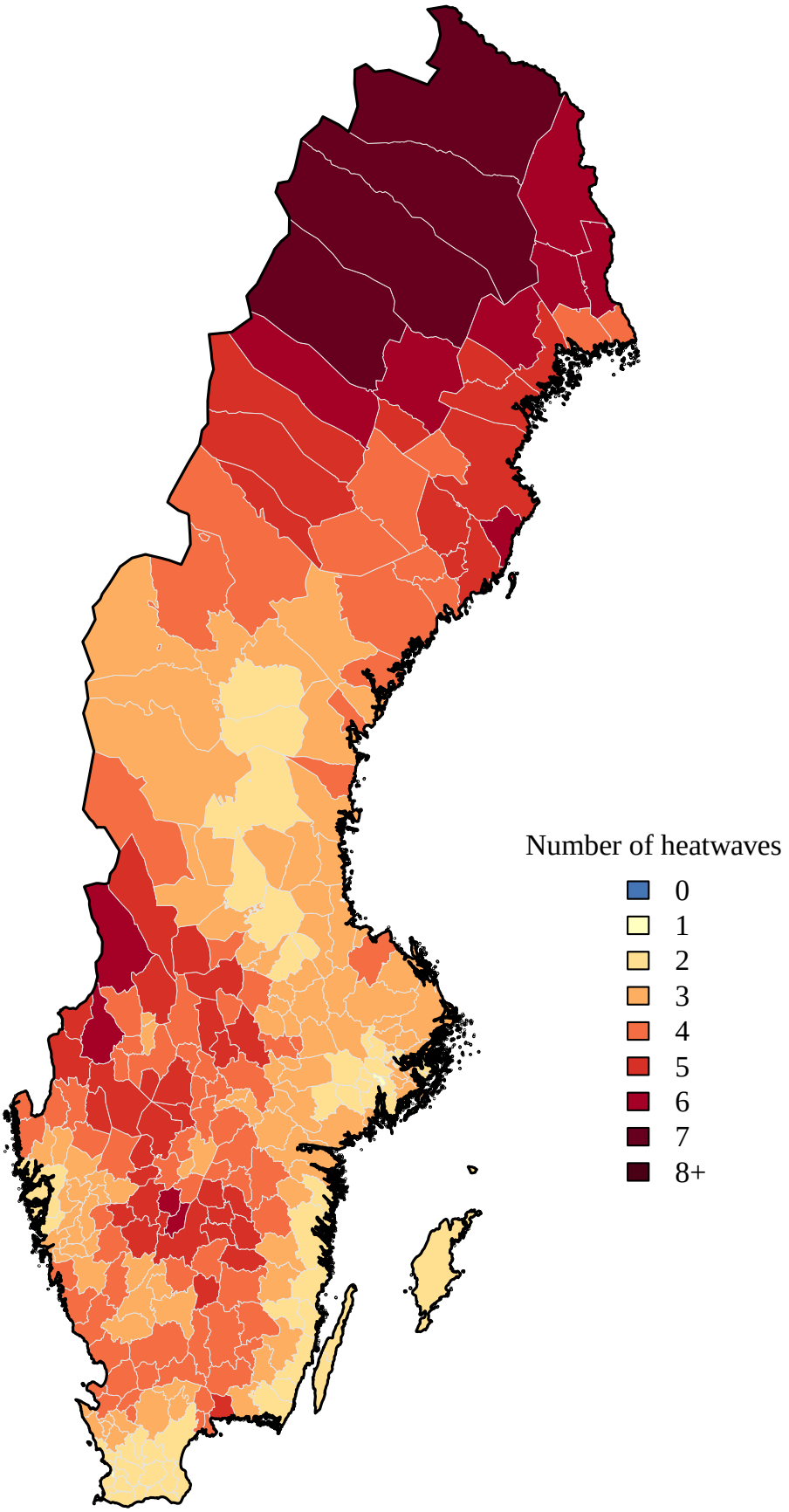
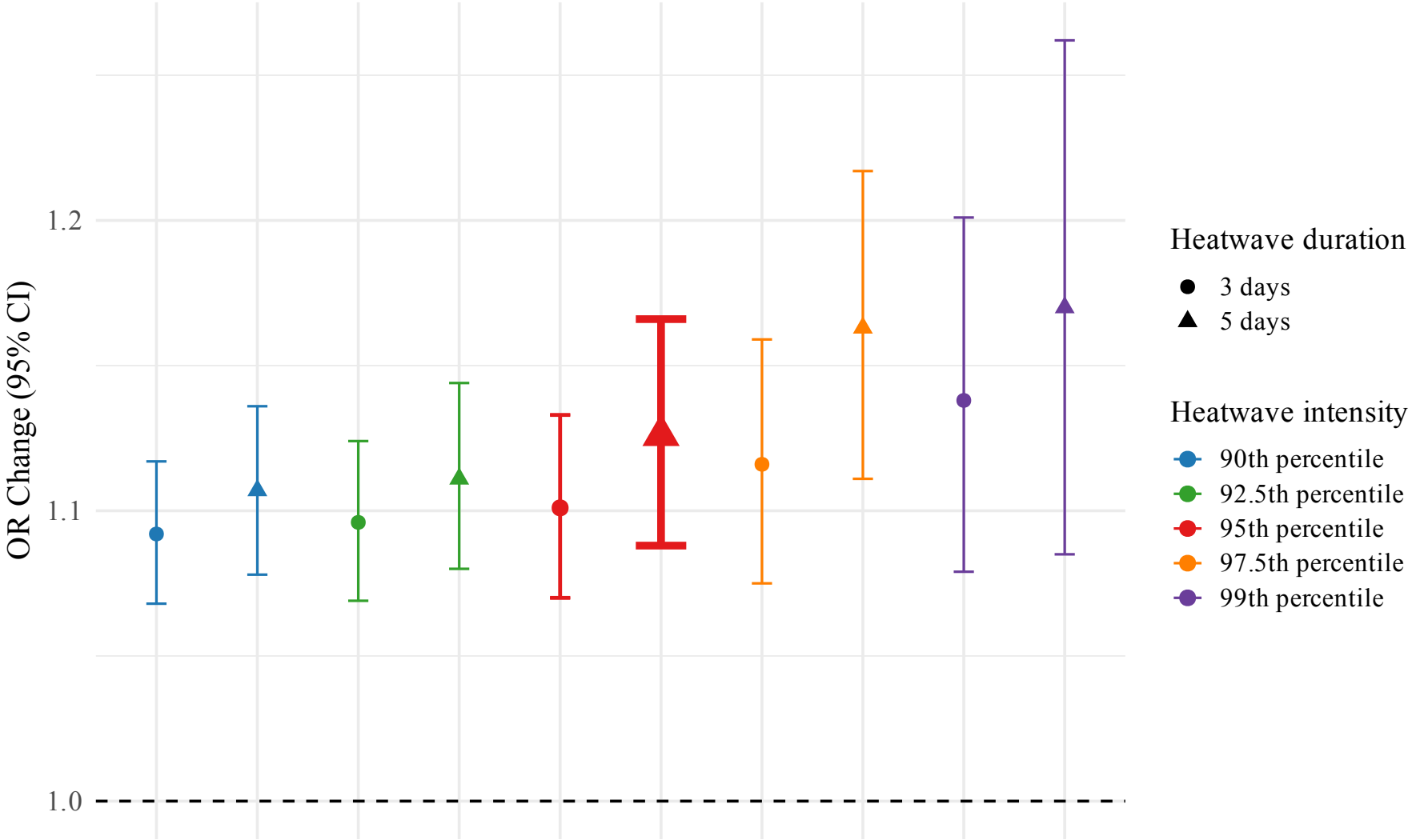


Figure 3: Sensitivity analyses



7 Tables

Table 1: Study population characteristics (N = 115,159).

Age (years)	
Median (IQR)	84 (78–89)
Mean (SD)	83.6 (7.5)
Sex, n (%)	
Female	59,058 (51.3%)
Male	56,101 (48.7%)
Education, n (%)	
Primary/lower secondary	59,914 (52.0%)
Upper secondary	38,384 (33.3%)
Post-secondary/tertiary	15,845 (13.8%)
Missing	1,016 (0.9%)
Income quintiles (SEK), n (%)	
1st quintile (0–133,900)	23,008 (20%)
2nd quintile (133,900–157,600)	23,008 (20%)
3rd quintile (157,600–229,700)	23,008 (20%)
4th quintile (229,700–306,600)	23,008 (20%)
5th quintile (306,600 or more)	23,009 (20%)
Missing	118 (0.1%)
Urbanicity status, n (%)	
Urban	85,840 (74.5%)
Rural	29,319 (25.5%)
Type of dwelling, n (%)	
Multi-family block	47,554 (41.3%)
Single-family house	41,577 (36.1%)
Residential home care	19,435 (16.9%)
Other housing	2,231 (1.9%)
Missing	4,362 (3.8%)
Building period, n (%)	
Before 1964	224,845 (39.1%)
1965–1974	118,560 (20.6%)
1975 or later	168,715 (29.3%)
Missing	63,675 (11.1%)

Note: Income quintiles were calculated among observations with non-missing income.

Table 2: Distribution of underlying causes of death (N = 115,159).

Underlying cause of death (ICD-10)	**n (%)**
Cancer	
All malignant neoplasms (C)	30,006 (26.1%)
Cardiovascular disease	
All cardiovascular disease (I)	42,034 (36.5%)
Coronary heart disease (I20–I25)	14,930 (13.0%)
Myocardial infarction (I21)	6,603 (5.7%)
Other heart disease (I30–I52)	12,252 (10.6%)
Stroke (I61, I62, I64)	1,345 (1.2%)
Respiratory disease	
All respiratory disease (J)	7,901 (6.9%)
Pneumonia (J12–J18)	2,291 (2.0%)
Chronic Obstructive Pulmonary Disease (J40–J44)	4,002 (3.5%)
Asthma (J45–J46)	143 (0.1%)
Metabolic disease	
Diabetes (E10–E14)	2,599 (2.3%)
All other causes	
All other ICD-10 diagnoses	32,619 (28.3%)

Table 3: Summary statistics of daily maximum temperature (°C) and spatial coverage of heatwaves across Swedish territory by year and month (May–September, 2014–18).

Year	Month	Min	Q1	Median	Mean	Q3	Max	IQR	% of Swedish territory covered by a p95 heatwave (≥5 days)
2014	May	−10.44	7.48	12.15	12.55	17.85	32.27	10.37	0.00
	June	−4.21	13.12	16.66	16.27	19.68	34.23	6.56	2.24
	July	2.71	20.95	24.02	23.65	26.77	38.73	5.82	98.99
	August	0.28	14.94	17.68	18.30	21.66	36.34	6.72	38.31
	September	−7.89	11.65	15.67	14.57	18.17	25.66	6.52	0.00
2015	May	−8.98	7.95	11.00	10.50	13.46	21.20	5.51	0.00
	June	−5.75	12.01	15.31	14.89	18.08	29.94	6.07	0.00
	July	−0.63	15.31	17.75	17.80	19.87	36.89	4.56	11.11
	August	1.40	17.47	20.17	19.70	22.57	30.41	5.10	27.13
	September	−1.58	11.85	14.38	14.05	16.47	25.54	4.62	0.00
2016	May	−5.73	10.66	14.63	14.54	18.25	30.63	7.59	0.00
	June	−4.69	14.87	18.60	17.94	21.38	34.83	6.51	0.11
	July	1.45	17.88	20.24	20.15	22.74	33.24	4.86	44.85
	August	−2.24	14.59	17.30	17.03	19.75	33.00	5.16	0.00
	September	−1.94	12.37	15.79	15.53	18.66	28.92	6.29	0.00
2017	May	−9.88	6.88	12.20	11.99	16.84	31.62	9.96	0.00
	June	−5.90	13.94	17.11	16.50	19.67	30.54	5.73	0.00
	July	−1.46	16.34	19.13	18.43	21.17	29.72	4.82	6.19
	August	−2.73	14.76	17.69	17.07	19.86	28.15	5.10	0.00
	September	−4.17	9.95	12.73	12.43	15.13	21.85	5.18	0.00
2018	May	−6.01	13.58	19.46	18.01	23.23	36.57	9.65	1.38
	June	−4.49	13.88	18.12	17.87	21.85	35.85	7.98	0.01
	July	1.96	22.52	25.25	24.85	27.79	37.05	5.27	98.94
	August	−1.09	15.17	18.36	18.68	21.99	36.49	6.81	7.21
	September	−7.08	10.89	14.38	13.90	17.24	27.35	6.35	0.00

Table 4: Odds ratios for all–cause mortality associated with heatwave exposure, stratified by sex and living environment characteristics (95% confidence intervals)

	OR (95% CI)
Total	**1.13 (1.08–1.17)**
Sex	
Female	**1.14 (1.08–1.19)**
Male	**1.11 (1.06–1.17)**
Education	
Primary/lower secondary	**1.09 (1.04–1.15)**
Upper secondary	**1.17 (1.10–1.24)**
Post-secondary/tertiary	**1.15 (1.04–1.26)**
Income quintiles (SEK)	
1st quintile (0–133,900)	**1.10 (1.02–1.8)**
2nd quintile (133,900–157,600)	**1.16 (1.07–1.25)**
3rd quintile (157,600–229,700)	**1.15 (1.07–1.25)**
4th quintile (229,700–306,600)	1.06 (0.98–1.14)
5th quintile (306,600 or more)	**1.18 (1.09 –1.28)**
Urbanicity status	
Urban	**1.14 (1.10–1.19)**
Rural	**1.08 (1.01–1.15)**
Type of dwelling	
Multi-family block	**1.09 (1.03–1.15)**
Single-family house	**1.12 (1.05–1.18)**
Residential home care	**1.18 (1.09–1.28)**
Other housing	1.21 (0.95–1.53)
Building period	
Before 1964	**1.08 (1.02–1.15)**
1965–1974	**1.11 (1.04–1.18)**
1975 or later	**1.12 (1.05–1.19)**

Bold = statistically significant (95% CI not including 1.00). OR = odds ratio

Table 5: Odds ratios for heatwave-related cause-specific mortality (95% confidence intervals), stratified by sex

Underlying cause of death (ICD-10)	OR (95% CI)	Females	Males
Cancer			
All malignant neoplasms (C)	**1.11 (1.04–1.19)**	1.08 (0.97–1.20)	**1.14 (1.04–1.25)**
Cardiovascular disease			
All cardiovascular disease (I)	**1.11 (1.05–1.18)**	**1.11 (1.03–1.20)**	**1.11 (1.02–1.21)**
Coronary heart disease (I20–I25)	**1.19 (1.08–1.31)**	**1.17 (1.01–1.35)**	**1.21 (1.06–1.38)**
Myocardial infarction (I21)	**1.24 (1.07–1.43)**	**1.27 (1.03–1.56)**	1.21 (1.00–1.48)
Other heart disease (I30–I52)	**1.14 (1.02–1.27)**	**1.18 (1.02–1.35)**	1.10 (0.93–1.29)
Stroke (I61, I62, I64)	0.90 (0.65–1.26)	0.95 (0.59–1.54)	0.86 (0.54–1.36)
Respiratory disease			
All respiratory disease (J)	**1.19 (1.04–1.37)**	**1.34 (1.11–1.62)**	1.05 (0.86–1.28)
Pneumonia (J12–J18)	1.24 (0.96–1.59)	**1.57 (1.11–2.23)**	0.97 (0.68–1.40)
Chronic obstructive pulmonary disease (J40–J44)	1.19 (0.98–1.44)	1.23 (0.95–1.60)	1.14 (0.86–1.51)
Asthma (J45–J46)	1.70 (0.72–4.01)	1.76 (0.67–4.61)	1.49 (0.22–10.00)
Metabolic disease			
Diabetes (E10–E14)	**1.26 (1.01–1.57)**	1.22 (0.89–1.67)	1.30 (0.95–1.79)

Bold = statistically significant (95% CI not including 1.00). OR = odds ratio from case-crossover

Appendix A. Popular Science Summary